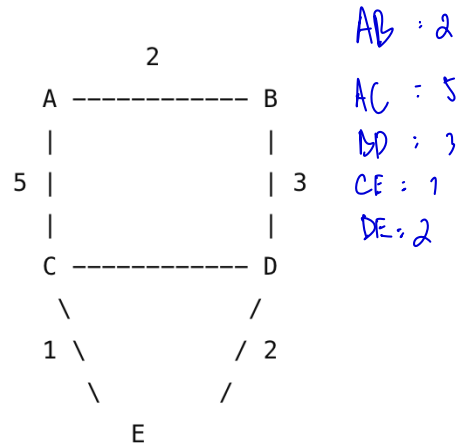


Exercise 2: Link State and Distance Vector Routing

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Learning objectives: Apply Dijkstra and Bellman-Ford; construct forwarding tables; analyze convergence, failures, and count-to-infinity.

Network topology:



Part 1 — Link State Routing

Question 1 — Dijkstra's Algorithm

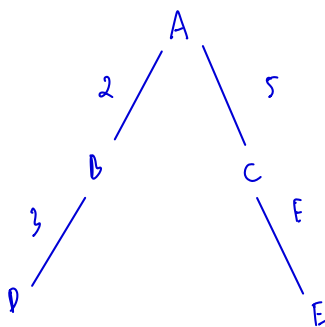
Starting at router A, execute Dijkstra's algorithm. Show every update and the node confirmed at each step.

Step	Confirmed Node	D(B), Prev.	D(C), Prev.	D(D), Prev.	D(E), Prev.
0	A	2, A	5, A	-	-
1	B	2, A	5, A	5, B	-
2	C	2, A	5, A	5, B	6, C
3	D	2, A	5, A	5, B	6, C
4	E	2, A	5, A	5, B	6, C

Question 2 — Shortest-Path Tree

Draw the shortest-path tree rooted at A, then list the shortest path and total cost.

SPT sketch:



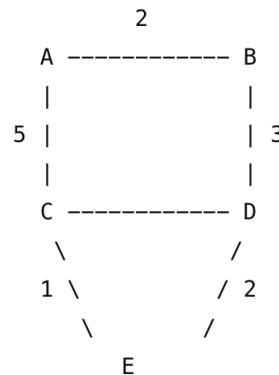
Destination	Shortest Path	Total Cost
B	A → B	2
C	A → C	5
D	A → B → D	5
E	A → C → E	6

Question 3 — Forwarding Table

Destination	Shortest Cost	Next Hop	Complete Path
B	2	B	A → B
C	5	C	A → C
D	5	B	A → B → D
E	6	C	A → C → E

Part 2 — Distance Vector Routing

Consider the same network. Initially, each router knows only the cost to its directly connected neighbors.



Question 4 — Bellman-Ford Update

Router A receives the following distance vectors from its neighbors:

From B:

A=2, B=0, C=4, D=3, E=5;

From C:

A=5, B=4, C=0, D=1, E=1.

Use the Bellman-Ford equation. Calculate router A's new distance to:

Show all calculations, including the neighbor through which the minimum-cost route is obtained.

Destination	Via B calculation	Via C calculation	Minimum Cost	Next Hop
B	$C(A,B) + D_B(B) = 2 + 0 = 2$	$C(A,C) + D_C(B) = 5 + 4 = 9$	2	B
C	$C(A,B) + C(C) = 2 + 4 = 6$	$C(A,C) + C(C) = 5 + 0 = 5$	5	C
D	$C(A,B) + D(D) = 2 + 3 = 5$	$C(A,C) + D(D) = 5 + 1 = 6$	5	B
E	$C(A,B) + D(E) = 2 + 5 = 7$	$C(A,C) + D(E) = 5 + 1 = 6$	6	C

Question 5 — Distance Vector Table

After receiving the distance vectors, construct router A's updated routing table.

Destination	Next Hop	Distance	Full Path
B	B	2	A → B
C	C	5	A → C
D	B	5	A → B → D
E	C	6	A → C → E

Part 3 — Routing Convergence

Question 6 — Link Failure

The C–D link fails. Explain separately what happens with Link State and Distance Vector routing.

Link State routing.

Your answer should describe:

- Which router(s) detect the failure.
- What information is flooded through the network.
- How routers update their link-state databases.
- How routers calculate new shortest paths.

Distance Vector routing.

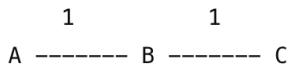
Your answer should describe:

- How routers learn about the failure.
- How updated distance vectors propagate.
- Why routers may temporarily have incorrect routing information.
- How routing eventually converges.

Link State	Distance Vector
Router C,D detect failure C-D via interface/Hello Message	Router C,D detect failure C-D via interface/Hello Message
C,D send Link State Advertisement (LSA) via network flooding	C,D send Distance Vector advertisement via network flooding
Router receives Link State DB via network flooding	Router receives Distance Vector advertisement via network flooding
Every router run Dijkstra's Algo on LSPB to calculate shortest path	Every router run Bellman-Ford algorithm to calculate shortest path

Part 4 — Count-to-Infinity

Consider



Initially A reaches C through B at cost 2. The B–C link fails, but B receives stale information from A saying C is reachable at cost 2.

Question 7

1. Why can this create a routing loop?

: In this scenario, Router B receives stale information from Router A saying C is reachable at cost 2. Router B then updates its own distance to C as 3 (cost to A + cost from A to C). Router A then receives this information and updates its distance to C as 4 (cost to B + cost from B to C). This process continues until the distance to C becomes infinity.

2. Show how the estimated distance to C can increase over successive iterations.

$c(B,C) : \infty$
 $c(A,C) : c(A,B) + c(B,C) : 1 + \infty \rightarrow \infty$
 $c(B,C) : c(B,A) + c(A,C) : 1 + \infty \rightarrow \infty$

3. What is count-to-infinity?

: The process where the distance to a destination increases over successive iterations until it reaches infinity.

4. Why does it occur in Distance Vector routing?

: Because Distance Vector routing uses hop count as the metric, and it takes time for the information to propagate back to the source.

5. Name a mitigation approach.

: Split Horizon, Reverse, Triggered Update/Hello, Hold-time timer

Part 5 — Compare LS and DV

Question 8

Characteristic	Link State	Distance Vector
Basic algorithm	Dijkstra's Algo	Bellman-Ford Algo
Information maintained	Topology structure all LSDB	Distance Vector
Information exchanged	Link state	Distance Vector
Algorithm	Global, Centralized	Iterative, Asynchronous & Distributed
Topology knowledge	Full topology	Partial topology
Routing updates	Full topology Flooding	Incremental
Convergence	$O(N^2 \log N)$	Parallel in network
Routing loops	Impossible / self-healing	Reconverge
Count-to-infinity	No	Yes
Example protocol	OSPF, IS-IS	RIP, EIGRP, BGP

Part 6 — Design Question

Question 9

You are designing a routing system for an enterprise with 500 routers requiring fast convergence, accurate topology knowledge, efficient routing, and hierarchical design. Choose LS or DV and justify your choice with at least three technical arguments. Your answer should discuss at least three advantages and/or disadvantages of your chosen approach.

Answer: 1) Fast Convergence: In Network, Link up & Link down, LSDB is updated & recomputed, so it converges fast.

2) Hierarchical Design (OSPF Area): 500 Routers can be divided into multiple areas, reducing the size of LSDB in each area, saving Memory/CPU.

3) Loop-free & No Count-to-inf: LSDB contains full topology, so it can calculate the shortest path, avoiding loops and count-to-infinity.

Disadvantage: 1) High Memory & CPU Overhead: Storing full LSDB in every router requires more resources.

2) Complex Config: Configuring LSDB is more complex than DV.

Challenge — Algorithmic Analysis

Question 10

For N routers and E links, compare LS and DV in information maintained, communication overhead, computation, convergence, and scalability. Explain why large-scale intra-domain routing generally favors Link State.

Information Maintained: LS: $O(N \times E)$ in Router, DV: $O(N \times L)$; L is the number of links.

Communication Overhead: LS: Flooding $O(N \times E)$, DV: $O(N)$.

Computation Complexity: LS: Dijkstra $O(N^2 \log N)$, DV: Iterative algorithm.

Convergence & Scalability: LS: No loop, fast convergence, large scale.

DV: Count-to-inf, routing loops, slow convergence.